

Alcohols, Phenols and Ethers

JEE Chemistry — Notes, Formula Sheet, Practice Problems & Diagram

1. Notes

- Alcohols ($R-OH$) have the hydroxyl group attached to an sp^3 carbon; classified as primary, secondary, or tertiary based on the carbon it's attached to.
- Phenols have the $-OH$ group directly attached to an aromatic ring; they are more acidic than alcohols because the phenoxide ion is stabilized by resonance with the ring.
- Ethers ($R-O-R'$) have an oxygen atom bonded to two alkyl/aryl groups; they are relatively unreactive and often used as solvents.
- Alcohols undergo oxidation: primary alcohols oxidize to aldehydes then carboxylic acids; secondary alcohols oxidize to ketones; tertiary alcohols resist oxidation (no H on the carbon bearing $-OH$).
- Lucas test distinguishes primary, secondary, and tertiary alcohols based on the rate of reaction with Lucas reagent ($ZnCl_2 + HCl$) - tertiary reacts instantly (turbidity), secondary reacts in a few minutes, primary reacts very slowly.
- Williamson ether synthesis: an alkoxide ion reacts with a primary alkyl halide in an SN_2 reaction to form an ether.
- Phenols undergo electrophilic aromatic substitution more readily than benzene because the $-OH$ group is a strong activating, ortho/para-directing group.



Alcohol



Phenol



Ether

Structures of an alcohol ($R-OH$), a phenol (aromatic $-OH$), and an ether ($R-O-R'$).

2. Formula Sheet

Alcohol general formula	$C_n H_{(2n+1)} OH$
Williamson ether synthesis	$R-O^- + R'-X \rightarrow R-O-R' + X^-$
Oxidation of primary alcohol	$R-CH_2OH \rightarrow R-CHO \rightarrow R-COOH$
Oxidation of secondary alcohol	$R_2CH-OH \rightarrow R_2C=O$ (ketone)
Dehydration of alcohol (to alkene)	$R-CH_2-CH_2-OH \xrightarrow{(H_2SO_4, \text{ heat})} R-CH=CH_2 + H_2O$
Acidity order (approx)	Carboxylic acid > Phenol > Water > Alcohol
Lucas reagent	$ZnCl_2 + \text{conc. HCl}$
Ether general formula	$R-O-R'$

3. Practice Problems (22 questions)

- Classify $CH_3-CH_2-CH(OH)-CH_3$ as a primary, secondary, or tertiary alcohol.
- Explain why phenol is more acidic than a simple alcohol like ethanol.
- State the product formed when a primary alcohol is oxidized with an excess of an oxidizing agent like acidified $KMnO_4$.
- State the reagents used in the Lucas test and explain how it distinguishes primary, secondary, and tertiary alcohols.
- Write the equation for the Williamson synthesis of diethyl ether from sodium ethoxide and ethyl bromide.
- Explain why tertiary alcohols cannot be oxidized to a carbonyl compound under normal conditions.
- State whether phenol undergoes electrophilic substitution more or less readily than benzene, and explain why.
- State the product of acid-catalyzed dehydration of ethanol using concentrated sulfuric acid and heat.
- Explain why ethers have much lower boiling points than alcohols of comparable molecular mass.
- Explain the directing effect (ortho/para or meta) of the $-OH$ group in electrophilic substitution reactions of phenol.
- State the type of reaction (SN_1 or SN_2) involved in the Williamson ether synthesis.
- Explain why phenol reacts with $NaOH$ to form a salt, but a simple alcohol like ethanol does not react appreciably with $NaOH$.
- State the product formed when a secondary alcohol is oxidized with an oxidizing agent.
- Explain why the $C-O$ bond in phenol is stronger (shorter) than the $C-O$ bond in an alcohol like cyclohexanol.
- State two common industrial or laboratory uses of ethers.
- Explain what is meant by dehydration of an alcohol and give the general reagent/condition used.

17. State the order of reactivity of primary, secondary, and tertiary alcohols with Lucas reagent, from fastest to slowest.
18. Explain why methanol and ethanol are miscible with water in all proportions, while higher alcohols are not.
19. State the product formed when phenol reacts with bromine water (without a catalyst).
20. Explain why diethyl ether was historically used as an anesthetic, in terms of its physical/chemical properties.
21. State the general formula for a saturated ether with two alkyl groups.
22. Explain why the boiling point of phenol is higher than that of toluene (similar molecular mass), despite both having similar non-polar aromatic rings.

Answer Key

1. Secondary alcohol (OH on a carbon attached to two other carbons)
2. Conceptual - the phenoxide ion (conjugate base) is stabilized by resonance delocalization into the aromatic ring, making phenol a stronger acid
3. Carboxylic acid (R-COOH)
4. Conceptual - Lucas reagent ($\text{ZnCl}_2 + \text{HCl}$): tertiary alcohols react instantly (turbidity), secondary in a few minutes, primary very slowly (no visible reaction at room temp)
5. $\text{CH}_3\text{CH}_2\text{-O- Na}^+ + \text{CH}_3\text{CH}_2\text{-Br} \rightarrow \text{CH}_3\text{CH}_2\text{-O-CH}_2\text{CH}_3 + \text{NaBr}$
6. Conceptual - tertiary alcohol carbon has no H atom attached directly to it, so it cannot lose an H to form a C=O bond via oxidation
7. More readily - the -OH group donates electron density into the ring via resonance, activating it toward electrophilic substitution
8. Ethylene ($\text{CH}_2=\text{CH}_2$) via dehydration, plus water
9. Conceptual - ethers cannot form hydrogen bonds with each other (no O-H), so their intermolecular forces are weaker than alcohols, giving lower boiling points
10. Conceptual - -OH is an ortho/para-directing, activating group due to resonance donation of its lone pair into the ring
11. $\text{S}_\text{N}2$ (backside attack of alkoxide nucleophile on primary alkyl halide)
12. Conceptual - phenol's resonance-stabilized conjugate base (phenoxide) makes it acidic enough to react with NaOH; ethanol's conjugate base is not stabilized, so it does not react appreciably
13. Ketone ($\text{R}_2\text{C=O}$)
14. Conceptual - phenol's C-O bond has partial double-bond character due to resonance with the ring's pi system, making it shorter and stronger than a simple alcohol's C-O single bond
15. Conceptual - e.g., diethyl ether as a solvent for Grignard reactions; historically as an anesthetic
16. Conceptual - dehydration removes a water molecule from the alcohol, typically using concentrated H_2SO_4 and heat, forming an alkene
17. Tertiary > secondary > primary (fastest to slowest with Lucas reagent)
18. Conceptual - small alcohols can hydrogen-bond extensively with water; as the hydrocarbon chain grows, the non-polar character dominates, reducing water miscibility
19. White precipitate of 2,4,6-tribromophenol forms (electrophilic substitution, no catalyst needed due to phenol's high reactivity)
20. Conceptual - diethyl ether is volatile and has anesthetic properties affecting the central nervous system, though it has since been replaced by safer modern anesthetics

21. $C_nH_{(2n+1)}-O-C_mH_{(2m+1)}$ (general ether formula)

22. Conceptual - phenol can hydrogen-bond (via -OH) with other phenol molecules, while toluene (non-polar, only van der Waals forces) cannot, giving phenol a much higher boiling point

Note: conceptual/derivation questions are marked accordingly in the answer key. Numerical answers are rounded for clarity — show full working in your own solutions.